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# Multi-class classification of brain tumors using optimized CNN and transfer learning techniques

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**Brain tumors are abnormal cell growths within the brain or central nervous system that disrupt normal brain function and can be life-threatening. Early detection and precise classification of brain tumors are critical for effective treatment and improved patient prognosis. In this research, a Convolutional Neural Networks model with four convolution blocks is proposed for automated brain tumor detection and classification. The proposed scheme is applied to four types of MRI images, i.e., glioma tumor, meningioma tumor, no tumor, and pituitary tumor. The proposed architecture is optimized using the Adam optimizer, and the model is trained to minimize cross-entropy loss, enhancing classification accuracy across all four categories. The proposed model is compared with the three transfer learning models namely ResNet50, VGG19 and DenseNet121. Out of these models the proposed CNN model has outperformed with the value of accuracy as 98.5%, followed by second best accuracy at ResNet50 with value as 93.2%. After that, the category-wise analysis is performed on the proposed CNN model. Category 0 is numbered as Glioma, Category 1 as Meningioma, Category 2 as Pituitary, and Category 3 as No_tumor. Category 3 has obtained the highest value of accuracy as 0.98, followed by 0.96, 0.89 and 0.86 in category 2, category 1 and category 0 respectively. This analysis improves diagnostic accuracy for brain tumors and has the potential to aid radiologists in clinical settings.**

**Keywords** Brain tumor, Deep learning healthy, Death rate, Magnetic resonance imaging, Medical images

Clinical diagnosis is a vital component of modern healthcare, particularly for disorders like brain cancer, which is among the deadliest in the world. The human brain, a highly complex organ, coordinates many cellular processes and functions as the central hub for the whole body. The unregulated proliferation of cells results in the development of brain tumors, which are among the most serious neurological disorders. Brain tumors are classified into the primary brain tumor that occurs from the cells inside the human brain, as these are mostly non-cancerous. They are approximately 70% tumor cells, designed with the ability to metastasize only within the brain. Although, secondary brain tumors arise due to cancer cells' circulation through the bloodstream. Gliomas are malignant tumors that instigate from glial cells, affecting the supportive tissues of the brain and spinal cord. In contrast, pituitary tumors develop in the pituitary gland, impacting hormonal balance and causing symptoms that vary widely depending on the hormone affected[1–3]. Brain tumors, with fatal malignant growths, affect persons enough to kill them from the youngest to elder ages[4]. The WHO 2021 CNS Tumor Classification puts tumors into four different categories, from less to worse (I to IV), the level of decreasing survival time, and aggressiveness of the tumor[5]. The early and exact portrayal and classification of brain tumors (BT) before they become complicated is the key to devising successful treatments for them. Different imaging techniques are used to localize the different types of brain tumors, and each of them has a particular objective. The widely used imaging techniques consist of positron emission tomography (PET), radiography, ultrasonographic (US) etc. MRI (magnetic resonance imaging) appears to be among the techniques that have shown high efficiency in both tumor detection and further treatment of cancer patients[6]. MRI is the main diagnostic tool for brain cancer classification because of its capability to develop high-resolution images and detailed information on the tumor[7]. This process could be confusing due to its complexity and requires a highly skilled radiologist in order to make correct interpretation[8]. Oncologists use visual examination of medical imaging to diagnose malignancies and identify the sites of tumors. However, the borders of tumors are often indistinct as a result of the presence of adjacent healthy tissues. Manual identification is both labor-intensive and sometimes results in inaccurate tumor

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diagnosis[9]. Automated tools for classifying brain tumors are progressively supplanting traditional detection methods. Several contemporary techniques have been devised to categorize brain cancers in MRI images[10]. Nevertheless, current methods for detecting brain cancer do not provide the robust prediction capabilities necessary for precise diagnosis. Although there are several obstacles to producing high-quality classification, such as a lack of pictures, inadequate image characteristics, insufficient data, and the imprecise location of the tumor, it is still possible to achieve great results in classification. It is indispensable that such a progressive and reliable CAD system based on deep learning concepts is put in place to fight the obstacles. This system is imperative since it becomes a particular diagnostic system used for patients for brain tumor treatments before and post-treatment in efforts to spur recovery.

Recent advances in deep learning have introduced cutting-edge technologies such as transfer learning models (e.g., ResNet, DenseNet, VGG) and hybrid architectures combining CNNs with SVMs or attention mechanisms. While these methods achieve competitive accuracy, they often require high computational resources, making them less suitable for real-time clinical use. To address this limitation, we propose an optimized lightweight CNN model that leverages modern training strategies (Adam optimizer, cross-entropy loss) to balance high accuracy with computational efficiency. This positions our work within the scope of cutting-edge research, while addressing gaps in clinical applicability.

The novelty of the proposed work lies in integrating modern CNN practices into a compact yet powerful architecture that surpasses several cutting-edge methods reported in literature, including VGG-16, CNN-SVM, and other transfer learning-based approaches. Unlike heavier models that demand extensive computational resources, our optimized CNN is computationally efficient and better suited for deployment in real-time healthcare applications.

The novelty of this work lies in the development of a computationally efficient CNN architecture specifically optimized for MRI-based brain tumor classification. By integrating adaptive feature extraction layers and lightweight design principles, the proposed model achieves high accuracy with reduced training time and resource requirements, distinguishing it from conventional deep CNN or transfer learning approaches. Figure 1 displays the study's proposed theme.

The remainder of the paper is organised as follows: Sect. Literature survey addresses the literature, Sect. Methodology presents the technique, Sect. Experimental evaluation and model comparison presents an ablation study, and Sect. Conclusion concludes the research.

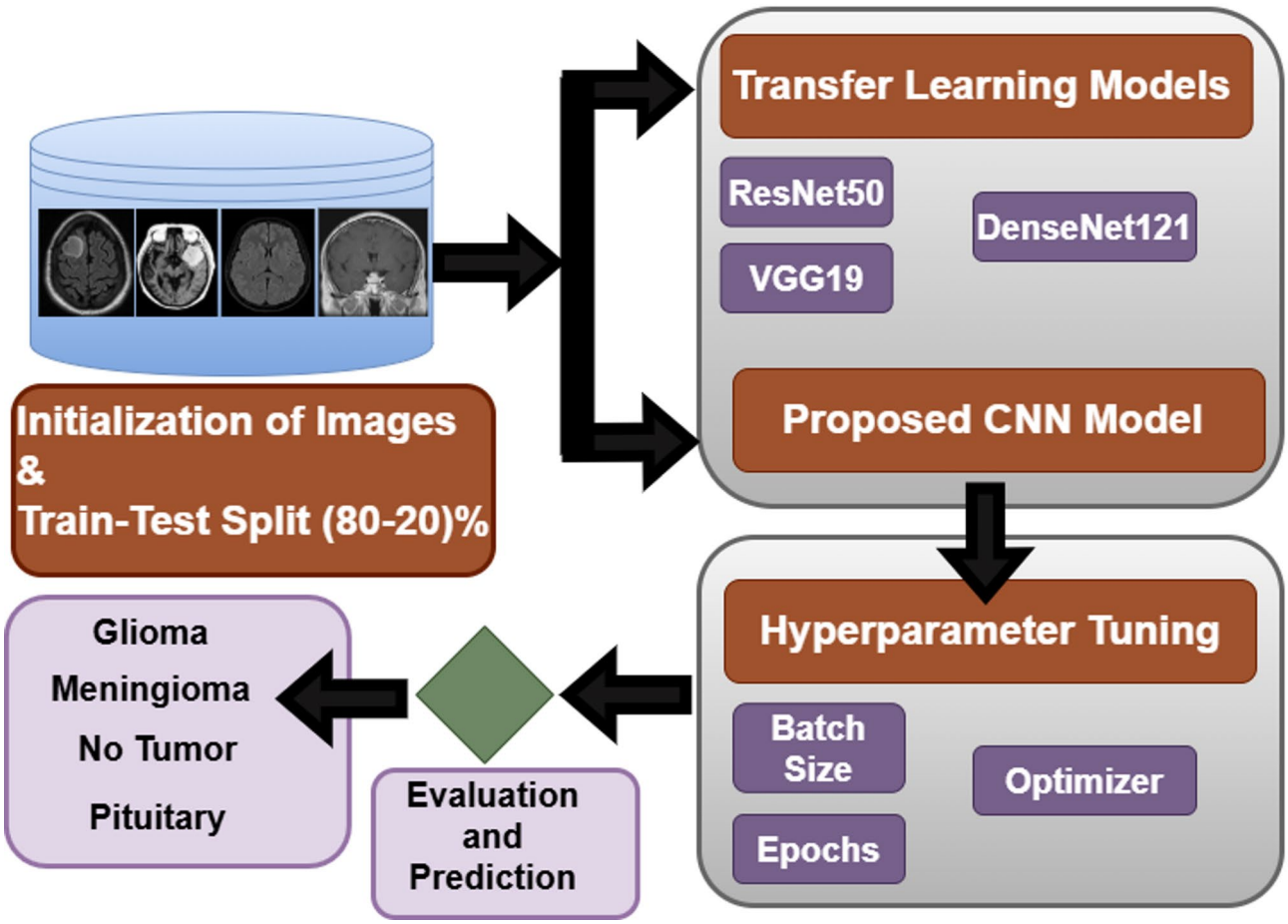



**Fig. 1**. Proposed Theme.

## Literature survey

The literature on classification of brain tumor with deep learning methods like CNN, LSTM, transfer learning models and VIT highlights the significance of early detection through techniques such as MRI scans. Annadurai et al.[9] designed a cascaded LSTM network and worked on a 2000 MRI image dataset. They obtained precision and accuracy of 89.5% and 98.5%, respectively. Nino et al.[10] worked on 3847 MRI images using ResNet50 architecture. They achieved the values of accuracy and precision at 92% and 94%. Yakub et al.[11] did a comparative analysis of different optimizers, and they worked using the BraTS2015 dataset. They obtained an accuracy value of 99.2%. Khaliki et al.[12] classified brain tumors using CNN and transfer learning-based models. They obtained an accuracy value of 98%. Zoghby et al.[13] focussed on the classification of three types of brain tumors using a dual convolutional neural network with VGG16 architecture. They worked on 3064 images and obtained a 99% accuracy value. Sivadas et al.[14] had done a 3-class classification of brain tumors. Their system obtained a value of accuracy of 98%. Srivastava et al.[15] presented brain tumor classification using the Br35H dataset. They worked on brain MRI images and obtained an accuracy of 81.11%.

In their performance analysis of several models, Jaidka et al.[16] reported accuracy values as 89% for different datasets. A deep learning-based automated technique for brain diagnosis from MRI data was presented by Srinivasan et al.[17]. It incorporates a Convolutional Recurrent Neural Network (CRNN) for classification, Fuzzy method for tumor segmentation, Co-occurrence Matrix for feature extraction, and filters for noise reduction. The model obtained 98.79% specificity, 98.17% accuracy, and 91.34% sensitivity using the REMBRANDT dataset. Using T1-weighted contrast-enhanced MRI images, a novel Convolutional Neural Network (CNN) architecture was presented by Alqudah et al.[18] for the categorization of brain tumors. The model achieves high accuracy, sensitivity, and specificity, with the best results on uncropped images (99% accuracy) due to retaining all pixels. Cropped images show slightly reduced accuracy (98.4%), while segmented images perform lowest (97.62%) due to loss of relevant pixel information. A deep learning framework utilizing CNN for the binary categorization of brain cancers as tumor or non-tumor using MRI images was reported by Bhardwaj et al.[19]. The dataset comprises 3,000 images, with preprocessing steps like normalization and downsampling for efficient computation. The proposed CNN model includes layers like convolution, pooling, fully connected, and dropout to enhance performance. It attains a classification accuracy of 99.4% and a test error rate of 0.03%, outperforming previous models.

Uysal et al.[20] had focussed on multiclass classification of brain tumors using deep learning models. It evaluates the performance of pre-trained models on two publicly available MRI datasets, representing normal tissue and three tumor types: meningioma, glioma, and pituitary adenoma. The study uses the Contrast Limited Adaptive Histogram Equalization (CLAHE) pre-processing technique to enhance image quality. Results show that RegNetY_16GF achieved the highest accuracy (96.52%) on one dataset, while ResNet50 performed best (99.43%) on the other. Anand et al.[21] explored ensembling of models using MRI images. Combining VGG19, and CNNs the model utilizes grid search to optimize weight combinations. It achieves superior accuracy (98.18%) compared to individual models by leveraging diverse feature spaces and mitigating overfitting. The dataset includes 3929 MRI images, with the ensemble model outperforming traditional approaches. Future research could include additional imaging modalities and segmentation techniques. Montaha et al.[22] designed a hybrid TimeDistributed-CNN-LSTM model for classifying brain tumors. Ablation studies were conducted to optimize architecture and hyperparameters, achieving 98.9% accuracy on the BraTS dataset. Comparative evaluation with a 3D CNN model highlights the proposed approach's superiority in precision, recall, and specificity.

Author[23] presented a method for the automatic grading of brain tumors applied to magnetic resonance spectroscopy (MRS) data. It extracts advanced features like spectral entropy and instantaneous frequency from MRS signals, achieving high accuracy (98.2%), sensitivity (100%), and specificity (97.53%) in classifying tumors into four grades based on WHO standards. Table 1 shows the related work findings and depicts existing methods and techniques name, outcome and limitations.

Author[26] presents a dual-phase deep learning pipeline for skin disease classification, addressing noise-related challenges in dermatological images. A cascaded U-Net first denoises images, preserving crucial lesion features, followed by a feature-selective CNN that employs Recursive Feature Elimination and feature fusion to enhance disease pattern recognition. Evaluations on standard datasets show improved classification accuracy over traditional methods. Anand et al.[27] proposes a fusion model integrating U-Net and CNN for early and accurate skin disease detection. U-Net segments dermoscopy images to extract regions of interest, while CNN performs multi-class classification on segmented images. Evaluated on the HAM10000 dataset using Adam and Adadelta optimizers, the model achieved 97.96% accuracy with Adadelta, outperforming existing state-of-the-art methods, demonstrating its effectiveness in automated skin disease diagnosis.

Belaid et al.[28] presented a deep learning–based framework for brain tumor classification using a combination of original MRI images and texture features. A dual-input VGG16 model is proposed, where original images and GLCM-based feature images (contrast and energy) are processed in parallel and their features concatenated. Experimental results show that combining original images with GLCM energy features yields superior performance, achieving an average accuracy of 96.5%. The approach outperforms existing state-of-the-art methods in classifying meningioma, glioma, and pituitary tumors. Huang et al.[29] proposed a convolutional neural network based on complex networks (CNNBCN) for automatic brain tumor classification from MRI images. The network architecture is generated using randomly constructed complex graphs and enhanced with a modified combination of activation functions. The proposed model achieves 95.49% classification accuracy and lower test loss than ResNet, DenseNet, and MobileNet. The study demonstrates that randomly generated network structures can outperform manually designed CNNs while offering a novel and efficient approach to neural network design. Sejuti et al.[30] proposed a hybrid CNN–SVM approach for brain tumor classification using MRI images. A custom 19-layer CNN is first used to extract discriminative features, followed by an SVM

| Ref/Year | Methods Used | Outcome | Limitations |
|---|---|---|---|
| [9] | Multiscale Deep CNN cascaded with LSTM | Accuracy = 89.5% | • Outputs depend on arbitrary mode compositions;<br>• Model must run across all possible combinations;<br>• Limited by dataset source |
| [10] | ResNet-50 architecture with added classification head | Accuracy = 92%<br>Precision = 94% | • Limited to binary classification (tumor vs. no tumor)<br>• Dataset size moderate (3847 images)<br>• Generalization to multi-class tumor types not tested |
| [11] | CNN architecture with comparative analysis of 10 optimizers | Adam Accuracy = 99.2% | • Limited computational resources restricted testing with multiple architectures<br>• Results specific to optimize performance on CNN, may not generalize across models<br>• NAG and RMSProp failures highlight instability |
| [12] | CNN-based transfer learning models (Inception-V3, EfficientNetB4, VGG19, VGG16) | VGG16<br>Accuracy = 98%,<br>F1-score = 97%,<br>AUC = 99%,<br>Recall = 98%,<br>Precision = 98%; | • Dataset skewed and of relatively poor quality<br>• Generalization may be limited to higher-quality or balanced datasets |
| [13] | Dual Convolutional Tumor Network (DCTN): VGG16 + custom CNN; | Robust classification of glioma, meningioma, and pituitary tumors | • Potential risk of overfitting (very high training accuracy);<br>• Validation limited to tested dataset;<br>• Glioma grade classification and auto-segmentation not yet implemented (future work) |
| [14] | Deep transfer learning using pre-trained GoogLeNet for feature extraction | Accuracy = 98% | • Misclassifications still occurred (analysed in study);<br>• Model performance not tested beyond 3 tumor classes;<br>• Generalization to larger or more diverse datasets not yet validated |
| [15] | Custom 2D CNN with integration of handcrafted features (LBP, DWT) | Accuracy = 81.11% (LBP), 94.11% (DWT), 94% (raw MRI with ResNet) | • Accuracy still lower with LBP features; Evaluation limited to Br35H dataset<br>• Generalization to other datasets/modalities (PET, SPECT, Biopsy) not tested |
| [16] | Machine learning classifiers (Logistic Regression, others); Hybrid ensemble technique; 10-fold cross-validation | Accuracy = 89% (small dataset), 87% (large dataset | • Manual feature extraction required (time-consuming, risk of poor selection)<br>• Performance limited compared to deep learning; Dataset size influenced results |
| [17] | Classification with Convolutional Recurrent Neural Network (CRNN); Dataset: REMBRANDT (2480 training, 620 testing) | Accuracy = 98.17%; Specificity = 91.34%; Sensitivity = 98.79%; | • Limited to binary classification (glioma vs. normal)<br>• Results dataset-specific (REMBRANDT only)<br>• Future work needed for multimodal tumor classification and grade evaluation |
| [19] | Fully automated deep learning CNN framework for brain tumor classification | Accuracy = 99.4%<br>Loss: 0.0030% | • Dataset details not specified<br>• Generalization to multi-class tumor types not mentioned<br>• Performance may depend on dataset quality and size |
| [21] | Weighted average ensemble of three models: VGG19 transfer learning, CNN without augmentation, CNN with augmentation; Dataset: TCGA lower-grade glioma (3929 MRI images, 110 patients) | Accuracy = 98%; Precision = 98.25%; F1-score = 98.5% | • Limited to lower-grade glioma dataset<br>• Findings may not generalize to other tumor types or cancer forms<br>• Segmentation not included—future work may use PSPNet, UNet, DeepLab, FPN for better region separation |
| [22] | TimeDistributed CNN-LSTM (TD-CNN-LSTM) for 3D MRI sequences | Accuracy = 98.90% | • Binary classification only (HGG vs. LGG)<br>• Computationally intensive;<br>• Generalization to other tumor types or multi-class classification not tested |
| [24] | Ensemble of two segmentation networks: U-Net + 3D CNN; Feature extraction with Gray Level Co-occurrence Matrix (GLCM) | Accuracy = 99.40% (whole tumor), 98.46% (enhanced tumor), 98.29% (tumor core)<br>Precision = 99.41%, 98.51%, 98.35%<br>F1-score = 99.4%, 98.29%, 98.46%<br>Sensitivity = 99.39%, 98.41%, 98.25% | • Computationally intensive<br>• Focused on segmentation only, not classification<br>• Generalization to other tumor types or modalities not evaluated |
| [25] | Ensemble of U-Net + 3D CNN; Feature extraction with GLCM | Precision: 99.4% (whole tumor), 98.5% (enhanced tumor), 98.35% (tumor core | • Focused on segmentation only, not classification;<br>• Limited evaluation of sensitivity, specificity, F-score (planned for future work);<br>• Computational complexity |

**Table 1**. Related work findings.

classifier to improve classification performance. The method classifies three tumor types—glioma, meningioma, and pituitary—using minimally preprocessed images. Experimental results show that the CNN–SVM model achieves a high accuracy of 97.1%, demonstrating robustness and effectiveness for medical image classification. Swati et al.[31] presented a transfer learning–based approach for multiclass brain tumor classification from CE-MRI images using block-wise fine-tuning of deep CNNs. A pre-trained VGG19 model is progressively fine-tuned to effectively learn both low-level and domain-specific MRI features from a small dataset. The proposed method requires minimal preprocessing and no handcrafted features, achieving an average accuracy of 94.82%. Experimental results show that it outperforms traditional machine learning and existing CNN-based methods on the CE-MRI benchmark dataset. Ayadi et al.[32] proposed an automated computer-aided diagnosis system for brain tumor classification using deep convolutional neural networks. A novel CNN architecture is designed to accurately classify different brain tumor types from MRI images, reducing reliance on time-consuming manual diagnosis. The model is evaluated on three benchmark datasets and demonstrates strong classification performance. Experimental results show that the proposed approach outperforms existing methods, supporting its effectiveness for early and accurate tumor diagnosis. Paul et al.[33] applied deep learning–based convolutional neural networks to classify brain tumors from CE-MRI images into meningioma, glioma, and pituitary types. Using axial MRI images from 233 patients, along with additional tumor-free images, the model achieves an

average classification accuracy of 91.43% under five-fold cross-validation. The study demonstrates a generalized, image-only approach that provides consistent per-image and per-patient predictions. Results highlight the potential of CNNs to improve automated brain tumor diagnosis and support future extensions using multi-plane MRI data. Surendran et al.[34] compared ResNet50V2 and DenseNet121 for meningioma tumor classification from brain MRI using transfer learning and data augmentation. Both models achieve near-perfect accuracy on clean data, with ResNet50V2 reaching 100% and DenseNet121 99.86%, and Grad-CAM confirms correct focus on tumor regions. However, introducing Gaussian noise causes a sharp drop in accuracy to nearly 55% for both models. The study highlights that despite strong performance and interpretability on clean data, robustness to noise remains a critical challenge for real-world clinical deployment. Rajendran et al.[24] presented an ensemble-based brain tumor segmentation framework using MRI images that combines GLCM-based feature extraction with deep learning models. A U-Net and a 3D CNN are used as two independent segmentation networks, and their outputs are fused to improve accuracy and completeness. The proposed ensemble achieves state-of-the-art performance, with mean accuracy up to 99.40% and F-scores above 98% for whole tumor, enhanced tumor, and tumor core regions. Results demonstrate that combining complementary networks significantly improves robustness and precision in glioma segmentation. Rajagopal et al.[25] proposed an automated brain tumor segmentation approach using MRI images that combines GLCM-based feature extraction with deep learning models. An ensemble of U-Net and 3D CNN architectures is employed to segment different tumor sub-regions, exploiting their complementary strengths. The combined predictions yield improved accuracy for tumor core, enhanced tumor, and whole tumor regions. Experimental results show state-of-the-art performance, achieving precision up to 99.4% on the validation set. Jemina et al.[35] presented an improved optimization-based machine learning framework (ImpOML) for brain tumor classification using MRI images. The method employs an Extended Sea-Horse–optimized Multi-Layer Perceptron (ExSH-MLP) to distinguish between healthy and tumor tissues, enhancing training efficiency and accuracy. Evaluated on the BraTS-2021 dataset, the proposed approach achieves a high classification accuracy of 98.5%. Comparative results demonstrate that the ImpOML model outperforms existing methods in terms of accuracy and effectiveness. Jayalakshmi et al.[36] presented a comprehensive review of Federated Learning (FL) and AI-based techniques for disease detection in healthcare. By analyzing 50 research studies, it evaluates existing FL-enabled models, datasets, and performance metrics while identifying key challenges and research gaps. The review highlights FL's role in privacy-preserving, distributed medical data analysis and its impact on improving diagnostic accuracy. Overall, the study provides insights and recommendations to guide the development of robust and efficient FL-based disease detection systems in healthcare.

The major offerings of the study are:

- For automated brain tumor identification and diagnosis, a customized CNN including four convolutional blocks is created, showing better performance than transfer learning models. This model is especially made to automatically identify and categorize brain cancers from MRI images.
- The specific use of the Adam optimizer and cross-entropy loss for this hybrid architecture ensures efficient and stable training, potentially setting a new benchmark for hyperparameter configurations in similar medical imaging tasks.
- The proposed CNN model attains an outstanding accuracy of 98.5%, outperforming well-established transfer learning models such as ResNet50 (93.2%), VGG19, and DenseNet121, ensuring reliable tumor classification.
- A detailed analysis of classification accuracy across different tumor types is performed, with No Tumor (0.98 accuracy) performing the best, followed by Pituitary (0.96), Meningioma (0.89), and Glioma (0.86), highlighting the model's efficacy in characterizing tumor categories.
- The research enhances diagnostic accuracy for brain tumor detection using MRI scans and provides a robust AI-assisted tool that can support radiologists in making precise clinical decisions, potentially improving patient outcomes.

## Methodology

In this section, Fig. 2 depicts a proposed framework for brain tumor classification. It incorporates convolutional blocks with ReLU activation. Transfer learning models are explored for improved performance. The model is assessed using metrics like accuracy, recall, F1-score, and precision. Hyperparameter tuning is employed to optimize model performance.

The use of four convolutional blocks was intentionally chosen to balance model complexity and computational efficiency. Brain tumor MRI images contain both global structural patterns and fine-grained textural variations; therefore, a moderately deep CNN is sufficient to capture hierarchical features without risking overfitting on a relatively limited dataset. The chosen kernel sizes (primarily $3\times3$) are widely established as optimal for extracting localized features and preserving spatial resolution in medical images. This configuration allows progressive feature refinement across layers while maintaining manageable computational cost. The proposed CNN architecture is inherently modular, where each component contributes to building a practical diagnostic tool. The convolutional blocks serve as feature extractors from MRI scans, while the optimizer and loss function enable efficient model training. Benchmarking against transfer learning models demonstrates the advantages of our lightweight design for deployment. Together, these components can be integrated into a clinical decision-support application that inputs MRI images and outputs reliable tumor classifications in real time. The quantitative problem formulation is as follows:

Let the dataset D is given in Eq. (1) as

$$D = \{(x_i,\ y_i)\}_{i=1}^{N} \tag{1}$$

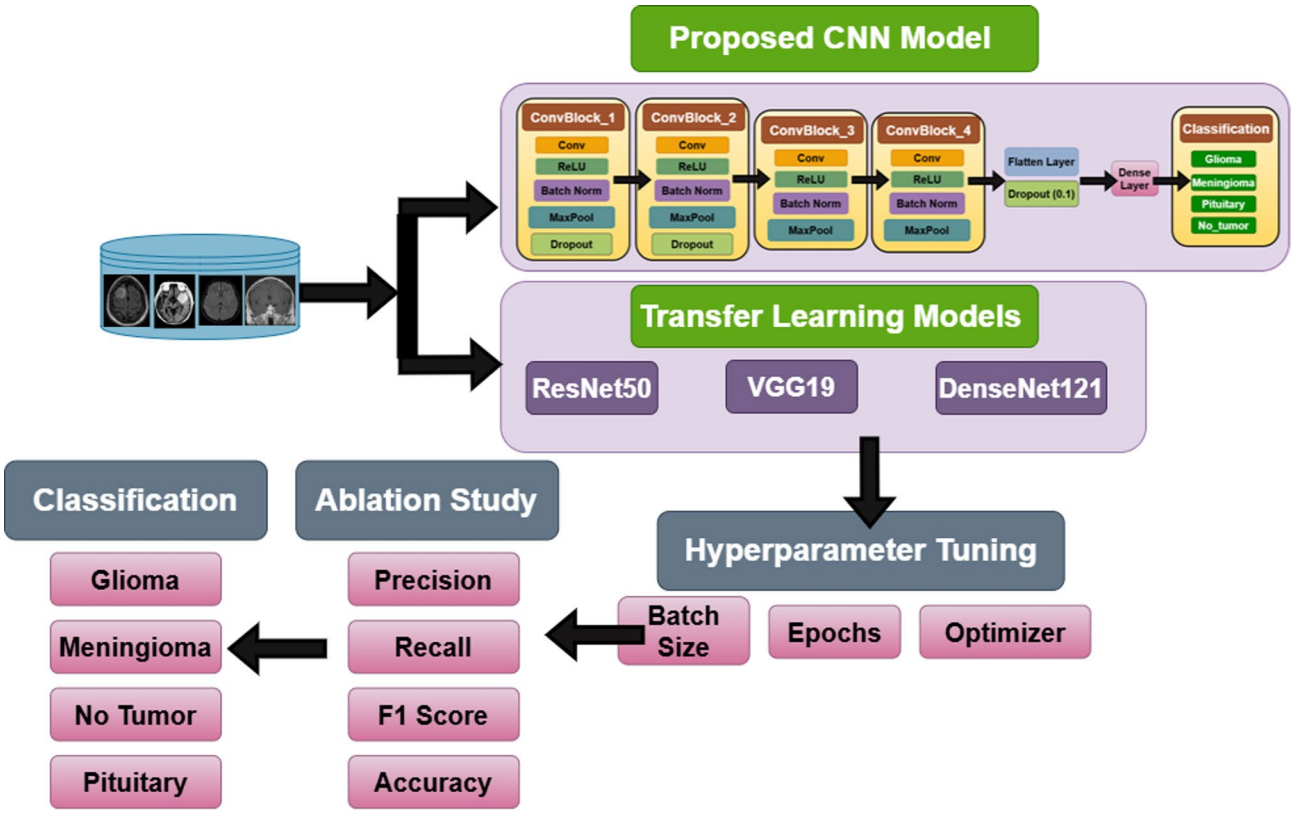



**Fig. 2**. Proposed Framework.

Where $x_i$ denotes the i-th MRI image and $y_i$ ε {0,1,2,3) denotes its label (0: Glioma, 1: Meningioma, 2: Pituitary, 3: No Tumor). The predicted class is depicted as shown in Eq. (2)

$$\widehat{y_i} = \arg max\, p_i,\ c \tag{2}$$

The training of the model is performed by minimizing the categorical cross-entropy over the training, the equation is given below in Eq. (3)

$$\pounds\ (\theta) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=0}^{3} y_i, c\, log p_i,\ c \tag{3}$$

$y_i, c$ is the one hot encoding of the true label, $p_i,\ c$ denotes the predicted probability that sample $x_i$ belongs to class c. Optimization is performed using the Adam optimizer with α as learning rate, $\beta_1$, $\beta_2$ and ε. The Adam update rule at timestep t is given in below Eqs. (4), (5).

$$M_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{4}$$

$$Vt = \beta_2 vt - 1 + (1 - \beta_2) gt \tag{5}$$

Here, $g_t = \nabla_\theta L(\theta_t)$ depicts gradient of loss at timestep t, $m_t$ depicts first moment estimate (exponentially decaying average of past gradients), $v_t$ depicts second moment estimate (exponentially decaying average of squared gradients), $\beta_1$, $\beta_2$ shows exponential decay rates for moment estimates, ϵ is the small constant for numerical stability.

**Pseudo Code**
**Input:** MRI Images (Glioma, Meningioma, Pituitary, No_Tumor)
**Output:** Predicted Tumor Class

**1. Load Dataset**
a. Split dataset into Train (80%) and Test (20%)
b. Assign class labels {0: Glioma, 1: Meningioma, 2: Pituitary, 3: No_Tumor}

**2. Preprocessing (Section 3.2)**
For each image:
- Resize image to 224 × 224
- Apply Image Normalization
- Apply Z-Score Normalization:
  img = (img − mean) / std

**3. Data Augmentation (Section 3.2)**
Apply the following transformations:
- Random rotation
- Horizontal/vertical flipping
- Scaling and zooming
- Width/height shifting

**4. Proposed CNN Architecture (Section 3.3)**
Initialize Model:
For block = 1 to 4:
  Conv2D(filters = f_block, kernel = 3×3, activation = ReLU)
  Batch Normalization
  MaxPooling2D(pool = 2×2)
Flatten Layer
Dense Layer (128 units, ReLU)
Dropout (rate = 0.5)
Output Layer (4 units, Softmax)

**5. Compile Model**
- Optimizer = Adam
- Loss = Categorical Cross-Entropy
- Metrics = Accuracy

**6. Training**
For epoch = 1 to N:
  Train model on augmented training data
  Compute training and validation loss
  Update weights using Adam optimizer
End For

**7. Evaluation**
- Evaluate model on test dataset
- Compute Precision, Recall, F1-score, AUC per class
- Generate Confusion Matrix (Figure 10)

**8. Explainability**
- Apply Grad-CAM to visualize activation regions (Figure 12)
- Confirm heatmaps focus on tumor areas

**9. Prediction**
For each new MRI image:
  Preprocess image
  Feed image to trained model
  Output predicted class (0–3)

End Algorithm

**Algorithm 1**. Proposed CNN-based brain tumor classification model.

## Dataset

The dataset comprises 3,264 MRI images of brain scans, organized into distinct training and testing folders for structured model development and evaluation[37]. Each of these folders is further divided into four subfolders, each representing one of the brain tumor classes: glioma, meningioma, pituitary tumor, and no tumor. Figure 3 illustrates all the image samples of different classes. This class-based organization ensures that the images are properly categorized, enabling streamlined data loading and pre-processing steps. This structured approach helps in maintaining consistency and facilitates robust model training.

Table 2 presents data on the distribution of four categories: No Tumor, Glioma, Pituitary, Meningioma. In the training set, there are 826 instances of Glioma, 822 of Meningioma, 827 of Pituitary, and 395 of No Tumor, totaling 2870 cases, which constitutes 88% of the overall data. The testing set includes 100 Glioma, 115 Meningioma, 74 Pituitary, and 105 No-Tumor cases, totaling 394, or 12% of the data. Overall, the dataset includes 926 Glioma, 937 Meningioma, 901 Pituitary, and 500 No-Tumor cases, making up 3264 instances.

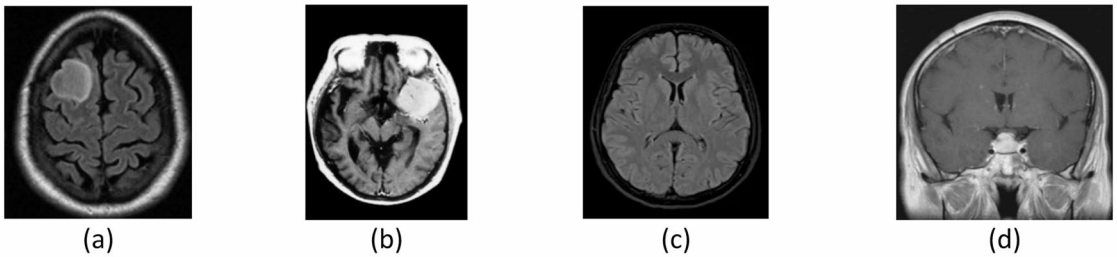



**Fig. 3**. Brain tumor samples: (**a**) Glioma_tumor, (**b**) Meningioma_tumor, (**c**) No_tumor, (**d**) Pituitary_tumor.

| No. of Images | Glioma | Meningioma | Pituitary | No_Tumor | Total | Distribution (%) |
|---|---|---|---|---|---|---|
| Train | 826 | 822 | 827 | 395 | 2870 | 88 |
| Test | 100 | 115 | 74 | 105 | 394 | 12 |
| Total | 926 | 937 | 901 | 500 | 3264 | 100 |

**Table 2**. Samples of brain tumor Dataset.

| Set | Glioma | Meningioma | Pituitary | No_Tumor | Total | Distribution (%) |
|---|---|---|---|---|---|---|
| Train | 1,770 | 1,790 | 1,720 | 880 | 6,160 | 88 |
| Test | 240 | 245 | 230 | 125 | 840 | 12 |
| Total | 2,010 | 2,035 | 1,950 | 1,005 | 7,000 | 100 |

**Table 3**. Dataset distribution after Augmentation.

## Preprocessing-augmentation pipeline

Prior to model training, several preprocessing operations were performed to standardize image quality and remove unwanted artifacts. Each dermoscopic image $I(x, y)$was resized to a fixed spatial resolution of $224 \times 224$ pixels to ensure compatibility with deep learning architectures.

(a) **Image Normalization**.

To maintain numerical stability during model optimization, pixel intensity values were normalized to a fixed range $[0,1]$ using min–max normalization shown in Eq. (6):

$$I_{norm}(x, y) = \frac{I(x, y) - I_{min}}{I_{max} - I_{min}} \tag{6}$$

where $I_{min}$ and $I_{max}$ denote the minimum and maximum intensity values of the image, respectively. This operation ensures uniform illumination and accelerates convergence during training.

(b) **Mean-Standard Deviation Normalization (Z-Score Normalization)**.

To reduce inter-image intensity variation, z-score normalization was applied as shown in Eq. (7):

$$I_z(x, y) = \frac{I(x, y) - \mu}{\sigma} \tag{7}$$

where $\mu$ and $\sigma$ represent the mean and standard deviation of pixel intensities across the training set. This step aligns all images to a standardized distribution with zero mean and unit variance.

Given the limited dataset size (3,264 dermoscopic images), data augmentation was employed to increase the diversity of training samples and reduce overfitting. Each original image $I(x, y)$was randomly transformed using a combination of geometric and photometric operations to generate synthetic variants, thereby improving the model's robustness and generalization.

(c) **Scaling and Zooming**.

To simulate variations in lesion size and camera distance, random scaling factors $s \in [0.8,1.2]$were applied as shown in Eq. (8):

$$T_{scale} = \begin{matrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{matrix} \tag{8}$$

(d) **Flipping**.

Horizontal and vertical flips were performed with a probability $p = 0.5$to model viewpoint variations as shown in Eq. (9):

$$T_{flip-x}(x, y) = (W - x - 1, y),\ T_{flip-y}(x, y) = (x,\ H - y - 1) \tag{9}$$

Here, W and H are the width and height.

(e) **Rotation**.

Images were randomly rotated within a range of $\theta \in [-20^\circ, +20^\circ]$using the rotation matrix. This preprocessing–augmentation pipeline not only standardized input quality but also effectively expanded the

training data by approximately sixfold. The resulting dataset improved model robustness, reduced overfitting, and ensured consistent performance across cross-validation folds.

After applying extensive data augmentation techniques, the total number of images in the dataset increased from 3,264 to 7,000 as shown in Table 3. The augmented dataset preserved the original class proportions across four tumor types — Glioma, Meningioma, Pituitary, and No_Tumor. As shown in Table X, the dataset was divided into 88% training and 12% testing sets to ensure balanced representation and reliable model evaluation. The augmentation process generated synthetic variations through random rotations, flips, brightness adjustments, zooming, and cropping, thereby enhancing the diversity of samples. This expansion not only improved the robustness and generalization capability of the model but also mitigated the risk of overfitting due to limited original data.

## Proposed CNN model

Figure 4 showcases a classic CNN architecture commonly used for image classification and similar tasks. It consists of two main components: convolutional Blocks and dense layers. Convolutional blocks are the workhorses of the network and are responsible for feature extraction. They typically consist of a sequence of convolutional layers, trailed by activation functions (ReLU), pooling layers, and potentially normalization layers. They are responsible for making the final classification or regression decisions. The model architecture consists of four convolutional blocks (convblocks), each designed to efficiently extract meaningful features from input images. Each convblock comprises multiple key components, including convolutional layers, ReLU activation layers, batch normalization, max-pooling layers, and dropout layers. Max-pooling layers help in reducing the spatial dimensions while preserving essential features, thus improving computational efficiency and reducing overfitting. Following the convolutional blocks, the network includes a flatten layer for classification followed by a fully connected dense layer, which refines the extracted features for better decision-making. Specifically, the convolutional and pooling layers, supported by batch normalization and dropout, form the feature extraction stage, responsible for learning hierarchical tumor-specific representations from MRI scans. The fully connected layers, followed by the softmax output, constitute the classification stage, which maps the extracted features into four tumor categories.

The proposed CNN consists of four convolutional blocks, each designed to progressively extract meaningful features from MRI scans.

- **Convolutional Layers**: These layers apply multiple filters to the input images to capture spatial hierarchies of tumor-related patterns such as edges, textures, and intensity variations. Early convolutional layers learn low-level features (e.g., edges and simple shapes), while deeper layers capture more abstract tumor-specific structures (e.g., irregular boundaries and heterogeneous textures).
- **Pooling Layers**: Max-pooling is used after convolutional layers to downsample feature maps, thereby reducing computational complexity and improving model generalization. This process retains the most salient features, helping the network focus on tumor-relevant regions while discarding redundant information.
- **Batch Normalization**: Applied after convolutional operations, batch normalization standardizes activations across mini-batches, which accelerates convergence and stabilizes training. In the context of brain tumor classification, this ensures that the network can effectively learn from MRI datasets of varied intensity distributions, thereby improving robustness.
- **Dropout Layers**: Dropout randomly deactivates a fraction of neurons during training, preventing the network from becoming overly dependent on specific feature pathways. This regularization strategy mitigates overfitting, which is a common issue in medical imaging tasks due to limited dataset sizes. For brain tumor classification, dropout enhances the generalization capability of the model, making predictions more reliable across unseen patient data.
- **Fully Connected Layers**: The final dense layers integrate the high-level features extracted by the convolutional blocks to perform classification. A softmax activation function is applied at the output layer to assign probabilities across the four classes: glioma, meningioma, pituitary tumor, and no tumor.

Finally, the classification layer categorizes the input data into four different classes. This structured approach enhances the model's capability to differentiate between various categories effectively.

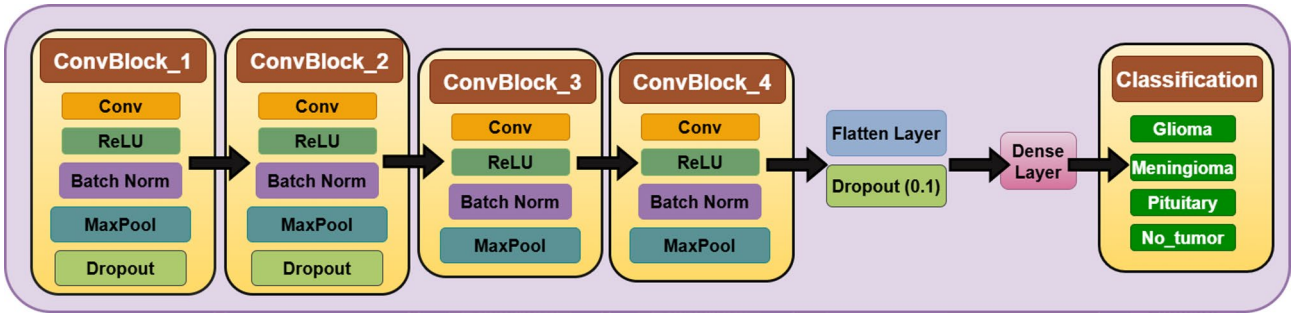



**Fig. 4**. Proposed CNN.

| CNN Architecture | Number of Trainable Parameters | Number of Layers |
|---|---|---|
| DenseNet121 | 7.98 Million | 121 |
| VGG19 | 143.67 Million | 19 |
| ResNet50 | 25.6 Million | 50 |

**Table 4**. CNN architectures used in study.

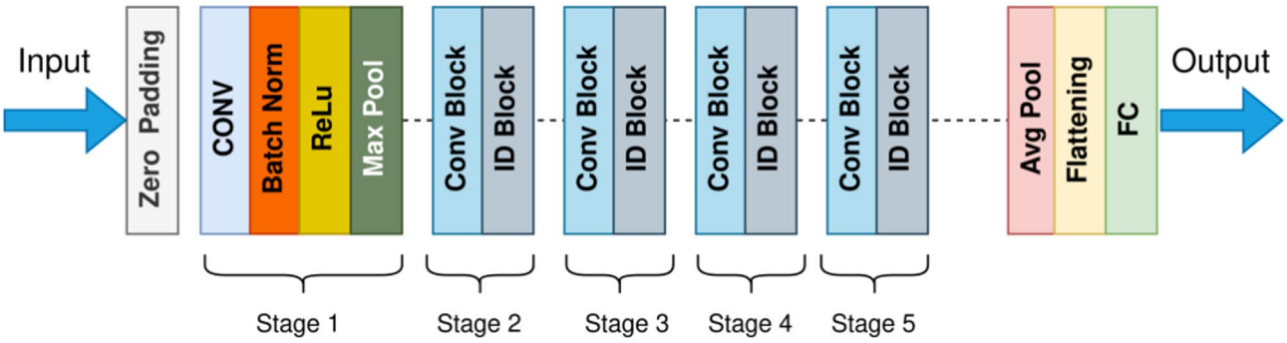



**Fig. 5**. ResNet50 architecture[39].

The proposed CNN employs four convolutional blocks to achieve an effective balance between feature extraction and computational efficiency. In preliminary experiments, shallower architectures (two or three blocks) resulted in insufficient feature representation and lower classification accuracy, while deeper architectures (five or more blocks) provided only marginal performance gains but substantially increased training complexity and the risk of overfitting, given the limited dataset size. Therefore, four convolutional blocks were selected as the optimal configuration. For convolutional operations, kernel sizes of $3 \times 3$ was adopted, as widely reported in prior literature such as VGG-based networks, owing to their ability to capture fine-grained local features with a reduced number of parameters. Larger kernels ($5 \times 5$ times, $7 \times 7$ times) were also tested but did not yield notable accuracy improvements and introduced higher computational costs. This empirically validated design ensures both strong classification performance and suitability for real-time applications. High-level spatial data are extracted from the MRI images by the CNN component, which enables the model to identify complex patterns and structural details that distinguish distinct tumor kinds.

The proposed CNN architecture introduces several modifications over conventional deep CNNs. Unlike standard models such as VGG-16 or ResNet, our framework employs a reduced number of convolutional layers optimized for lightweight computation, an adaptive kernel configuration that balances feature extraction and model complexity, and a hybrid activation and normalization pipeline (e.g., ReLU + Batch Normalization + Dropout) designed to prevent overfitting on limited MRI datasets. Furthermore, the use of a fine-tuned Adam optimizer with a dynamic learning rate schedule, and targeted data augmentation strategies enhances generalization while maintaining computational efficiency. These elements collectively constitute the novelty of our approach, making it suitable for real-time clinical use.

## Transfer learning models

In this study, three transfer learning models ResNet50, DenseNet121, and VGG19 are employed for automated brain MRI classification. Transfer learning leverages models pretrained on large-scale natural image datasets such as ImageNet, which contains over 14 million images across 1,000 categories. By utilizing pretrained weights, these models can extract rich hierarchical features from input images, even when the target dataset is relatively small, as is the case with brain MRI scans.

For adaptation to the four-class brain tumor classification problem (glioma, meningioma, pituitary, and no tumor), the original output layers of these models were replaced with a fully connected layer consisting of four neurons and a softmax activation function, enabling multiclass classification. The convolutional base of each model was optionally fine-tuned while training the top layers to optimize feature extraction specifically for MRI images.

All models were trained for 40 epochs, and their architectures, including the number of trainable parameters and layers, are summarized in Table 4. Performance evaluation was conducted using accuracy, precision, recall, and F1-score, providing a comprehensive assessment of each model's ability to extract relevant features and classify brain tumor types effectively.

*ResNet50*

ResNet-50 known for its efficiency in image classification and computer vision tasks as shown in Fig. 5. It has 50 layers and uses residual learning[38]. Introduced by Microsoft in 2015, ResNet-50 employs skip connections (shortcut paths) that help preserve gradients during backpropagation, allowing for easier training of deep

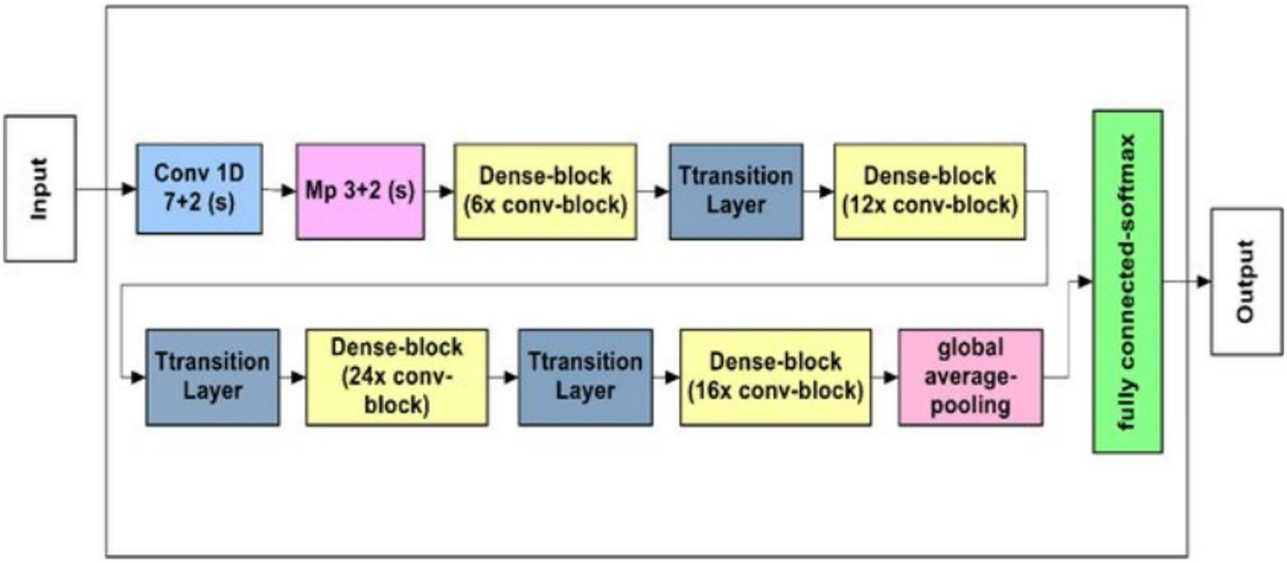



**Fig. 6**. DenseNet121 architecture[41].

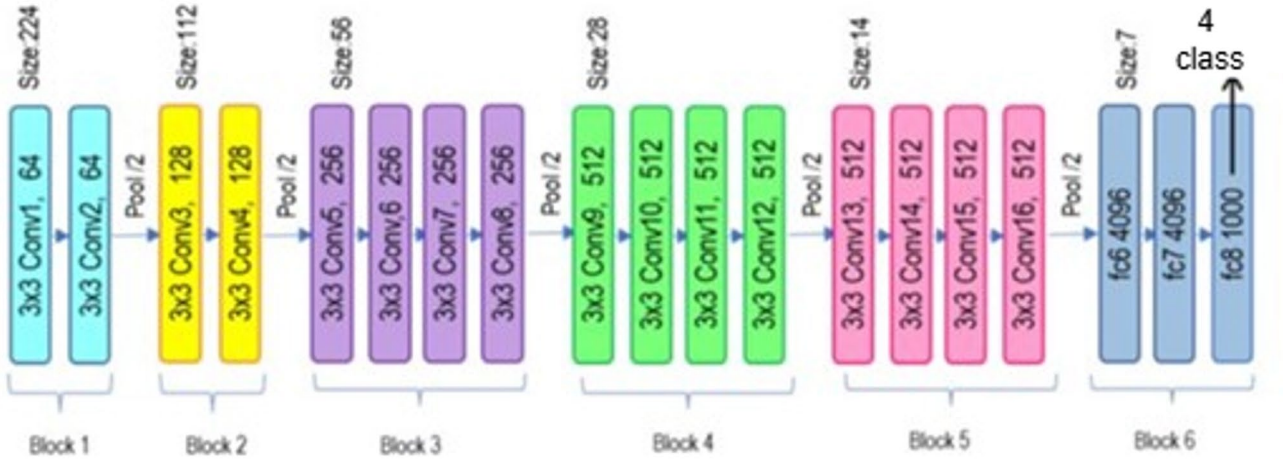



**Fig. 7**. VGG19 architecture[43].

networks. It is widely used in applications like object detection, medical imaging, and facial recognition. With 23 million parameters, ResNet-50 balances depth and efficiency, making it a popular choice for various deep learning tasks.

*DenseNet121*

DenseNet-121 (Dense Convolutional Network-121) is a deep learning architecture designed to enhance feature reuse and efficiency in image processing tasks as shown in Fig. 6. It consists of 121 layers and employs dense connections[40]. This connectivity improves gradient flow, reduces redundancy, and enhances learning efficiency. Introduced in 2017, DenseNet-121 requires fewer parameters than traditional deep networks like ResNet-50, making it computationally efficient. It is widely used in object identification, medical imaging, and image categorization. Due to its compact architecture and strong feature propagation, DenseNet-121 is a popular choice for deep learning applications.

*VGG19*

With 19 layers—16 convolutional layers and 3 fully connected layers—the VGG-19 is a deep convolutional neural network (CNN) renowned for its straightforward but efficient design[42]. VGG-19 uses small 3 × 3 convolutional filters, stacked sequentially, to capture complex image features. Despite its depth, it has a huge

quantity of parameters (~ 143 million), making it computationally expensive as shown in Fig. 7. It is widely used in image classification, object detection, and style transfer.

## Hyperparameter tuning

In order to optimize machine learning models and guarantee optimal performance, hyperparameter tuning is an essential step. Hyperparameters are preset settings that affect learning, as opposed to model parameters, which are learned during training[44]. More advanced methods, like Bayesian optimization, leverage probabilistic models to guide the search more efficiently. Proper hyperparameter tuning can significantly enhance a model's accuracy, generalization, and robustness. Without it, models risk underfitting or overfitting, leading to suboptimal performance. Therefore, selecting and fine-tuning hyperparameters is an essential process in developing high-performing machine learning models.

The quantity of training samples processed before to the model's weights being updated is referred to as the batch size. It is essential for model performance, memory utilization, and training effectiveness. Smaller batch sizes (e.g., 16 or 32) provide more frequent updates, leading to faster convergence but increased noise in gradients. Larger batch sizes (e.g., 128 or 256) offer more stable gradients but require more memory and may lead to slower generalization[45]. The optimal batch size is often determined empirically, balancing computational efficiency and model accuracy. Common choices follow powers of two for hardware optimization. Batch size is often selected as a power of two for efficient computation as shown in Eq. (10):

$$B = 2^n \text{n?}\{4,\ 5,\ 6,\ 10\} \tag{10}$$

Common values include 16, 32, 64, 128, and 256.

One whole run of the training dataset through the model is referred to as an epoch. In order to minimize the loss function, the model modifies its weights at each epoch in accordance with the calculated gradients. The model can learn better representations with more epochs, but if there are too many, it may overfit[46]. The optimal number of epochs is often determined using techniques like early stopping, where training halts once validation performance stops improving. Generally, models require multiple epochs to achieve convergence, but the exact number depends on the dataset, model complexity, and chosen optimization algorithm. Each epoch shows multiple iterations and Per Epoch Iterations are shown in Eq. (11), followed by other iteration Eqs. (12) and (13) below:

$$\text{Per epoch Iterations} = \frac{Training\ Samples\ (TS)}{Batch\ Size\ (BS)} \tag{11}$$

$$\text{Total Iterations} = \text{Epochs (E)} * \frac{\text{TS}}{\text{BS}} \tag{12}$$

$$\text{Epoch Completion Time} = T_{epoch} = \frac{TS}{BS} * T_{batch} \tag{13}$$

The model makes predictions using initial weights. The loss function calculates how far the predictions are from the actual values.

Adam (Adaptive Moment Estimation) is a popular deep learning optimization technique that combines the advantages of the RMSprop optimizer and Stochastic Gradient Descent (SGD) with momentum. It works especially well with noisy datasets, non-stationary targets, and sparse gradients. Equation (14) illustrates how Adam dynamically modifies the learning rate for every parameter depending on first- and second-order moment estimates.

$$mg_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{14}$$

$mg_t$ is the mean of gradients, $g_t$ is the gradient at time step 't'. $\beta_1$ $\beta_2$ is the decay rate that is typically 0.9, $v_t$ is the variance of gradients as shown in Eq. (15).

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \tag{15}$$

Table 5 depicts the Hyperparameters used with the values and justification of all of them.

## Experimental evaluation and model comparison

The ablation study analyzes model performance through accuracy and loss plots, showing trends in training and validation. A confusion matrix supported by key parameters and metrics evaluates classification effectiveness across classes, learning rate curves. Visualizations of results illustrate the model's predictive accuracy and robustness in distinguishing tumor types. Table 6 shows the hyperparameter details related to CPU, GPU, RAM etc.

The model was trained and evaluated on a system equipped with an Intel Core i5 processor, Intel Iris Graphics, and 16 GB RAM. The implementation was carried out using Anaconda Navigator and Spyder IDE for model development, training, and evaluation. The dataset was divided using an 80–20% train–test split to ensure balanced representation across all four classes. A total of 40 epochs and a batch size of 32 is used in the study. As detailed in Sect. 3.2, the preprocessing pipeline included image normalization, mean–standard deviation (Z-score) normalization, and geometric augmentations such as scaling, zooming, flipping, and

| Hyperparameter | Value | Justification |
|---|---|---|
| Number of Conv Blocks | 4 | Balanced depth for feature extraction and efficiency |
| Kernel/Filter Size | 3 × 33 \times 3 | Standard size to capture fine local features with manageable parameter count |
| Activation Function | ReLU (hidden layers), Softmax (output) | ReLU prevents vanishing gradients; Softmax for multi-class probability output |
| Pooling | Max Pooling (2 × 22 \times 2) | Dimensionality reduction while retaining dominant features |
| Batch Normalization | Applied after convolutional layers | Stabilizes learning and accelerates convergence |
| Dropout Rate | 0.5 (after fully connected layers) | Mitigates overfitting, especially with limited dataset size |
| Optimizer | Adam | Adaptive learning rate, efficient convergence |
| Learning Rate | 0.001 | Empirically determined optimal value |
| Loss Function | Categorical Cross-Entropy | Standard for multi-class classification tasks |
| Batch Size | 32 | Balanced memory use and gradient stability |
| Epochs | 50 | Sufficient for convergence without overfitting |

**Table 5**. Hyperparameter Justification.

| Item | Specification |
|---|---|
| CPU | Intel Core i5 |
| GPU / Accelerator | Intel Iris Graphics (integrated) |
| RAM | 16 GB |
| Operating System | Windows 10 |
| Python version | 3.10.6 |
| Deep learning framework | PyTorch 2.0.1, TensorFlow 2.12.0] |
| Key libraries | NumPy, scikit-learn, torchvision/keras |
| Random seed(s) | 42 |
| Training regime | Batch size = 32, Epochs = 40, Optimizer = Adam |
| Input image size | 224 × 224 |
| Augmentations | Scaling, Zooming, Rotation, Flipping |
| Average inference time | 65 ms |

**Table 6**. Hyperparameter Details.

| Models | Precision | Recall | F1-Score | Accuracy |
|---|---|---|---|---|
| ResNet50 | 86.5 | 88.7 | 94.3 | 93.2 |
| DenseNet121 | 87.6 | 86.9 | 83.2 | 89.1 |
| VGG19 | 72.4 | 75.6 | 73.5 | 75.0 |
| Proposed CNN | **96.2** | **89.7** | **92.2** | **98.5** |

**Table 7**. Ablation study of different models used in study.

rotation to enhance model generalization and reduce overfitting. Comprehensive evaluation metrics—including class-wise precision, recall, F1-score, and AUC—were reported for all models. Table 7 presents the precision, recall, F1-score, and accuracy values, while Fig. 10 illustrates the confusion matrix of the proposed CNN model, demonstrating its strong discriminative capability. Additionally, Fig. 12 provides Grad-CAM visualizations that highlight the model's ability to focus on clinically relevant tumor regions. Section 4.3 further discusses the performance of the proposed model across different datasets and imaging categories.

To assess computational efficiency, both the parameter count and inference speed were evaluated on a system equipped with an Intel Core i5 processor, Intel Iris Graphics, and 16 GB RAM. The lightweight architecture results in a modest number of trainable parameters, enabling efficient execution. The model achieves an average inference time of approximately 65 ms per image, indicating its suitability for near real-time diagnostic support in clinical workflows.

Figure 8 demonstrates the learning rate curve for all the models. The amount that the model changes at each stage is known as the learning rate. It regulates the model learning rate, which determines how quickly or slowly a model learns. A graph between the potential learning rate and validation loss values has been created in order to determine the learning rate for the top-performing model. The suggested CNN model with batch size 32 and Adam optimization yields the optimum learning rate curve.

It is clear from the learning rate curve and the confusion matrix parameters that the proposed CNN model has shown enhanced behaviour than other models. Therefore, in further sub-sections analysis will focus on the

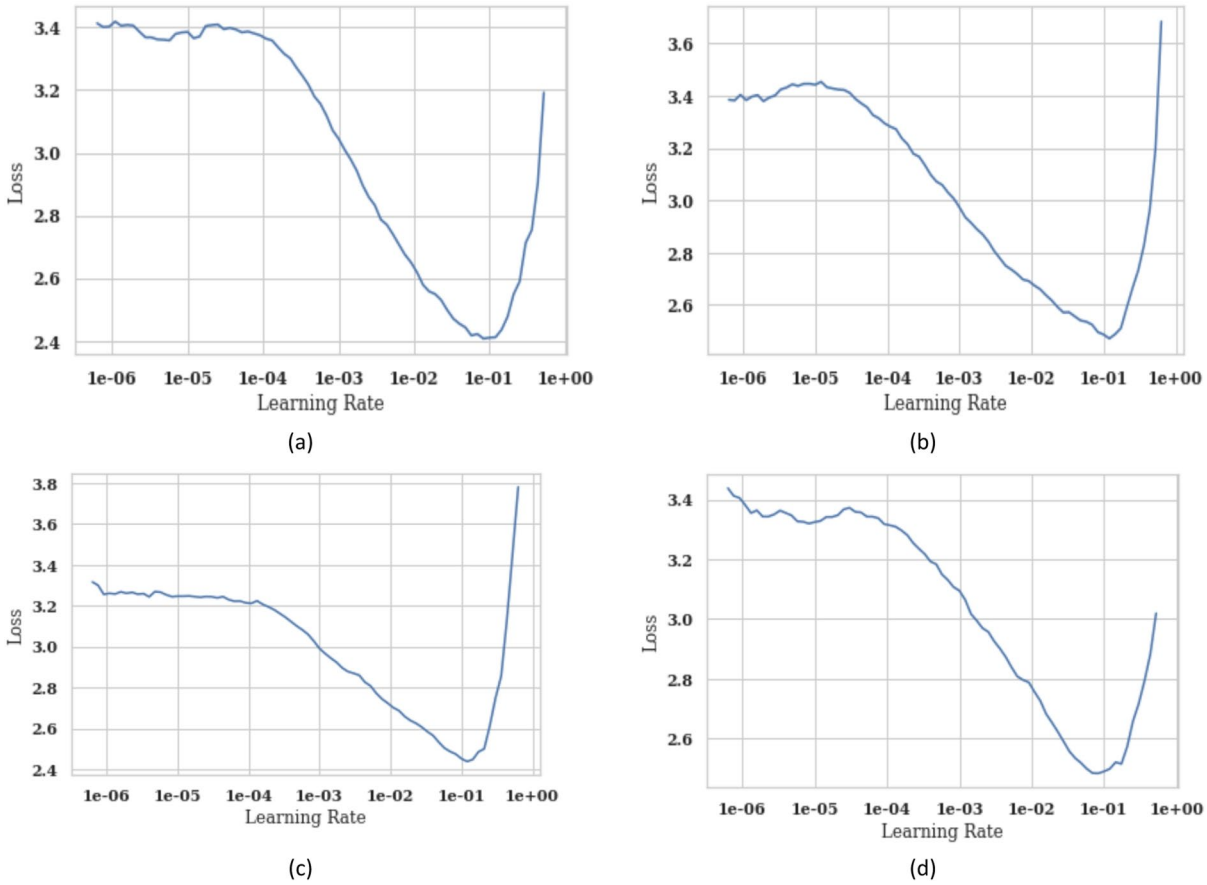



**Fig. 8**. Learning Rate (**a**) DenseNet121, (**b**) VGG19, (**c**) Proposed CNN, (**d**) ResNet50.

| Epochs | Val_Accuracy | Train_Accuracy | Loss | Train_Loss |
|---|---|---|---|---|
| 5 | 0.9021 | 0.90 | 0.2752 | 0.23 |
| 10 | 0.9313 | 0.92 | 0.1896 | 0.185 |
| 15 | 0.9554 | 0.94 | 0.1209 | 0.11 |
| 20 | 0.9805 | 0.97 | 0.0579 | 0.04 |
| 25 | 0.9801 | 0.975 | 0.0612 | 0.05 |
| 30 | 0.9875 | 0.978 | 0.0381 | 0.035 |
| 35 | 0.9938 | 0.987 | 0.0222 | 0.021 |
| 40 | 0.9856 | 0.98 | 0.0440 | 0.042 |

**Table 8**. Analysis of proposed CNN model at 40 Epochs.

proposed CNN to gain deeper insights into its performance. A category-wise evaluation will be conducted to assess how well the model classifies different categories.

## Accuracy and loss curves of proposed CNN model

Throughout each training session, substantial improvements in the model's performance are seen, as shown in Table 8; Fig. 9 (a) and (b). During Epoch 5, the model attained an accuracy of roughly 90.21%. Nevertheless, there was still potential for improvement since the loss, which quantifies the difference between the anticipated and actual values, remained at about 0.2752. Regarding Epoch 10, there was a significant improvement in accuracy, reaching around 93.13%. Similarly, the loss decreased to about 0.1896, suggesting that the model's forecasts were becoming more accurate. At Epoch 15, the model's accuracy increased to about 95.54%, demonstrating its enhanced capability to produce very precise predictions.

During Epoch 20, the model reached a noteworthy milestone by obtaining an outstanding accuracy of about 98.05%. This significant enhancement showcased the efficacy of the training procedure. The loss further dropped to around 0.0579. By the time the training reached Epoch 25, the model consistently maintained a high level of accuracy, remaining close to 98.01%. Despite a small rise in the loss to around 0.0612, it remained

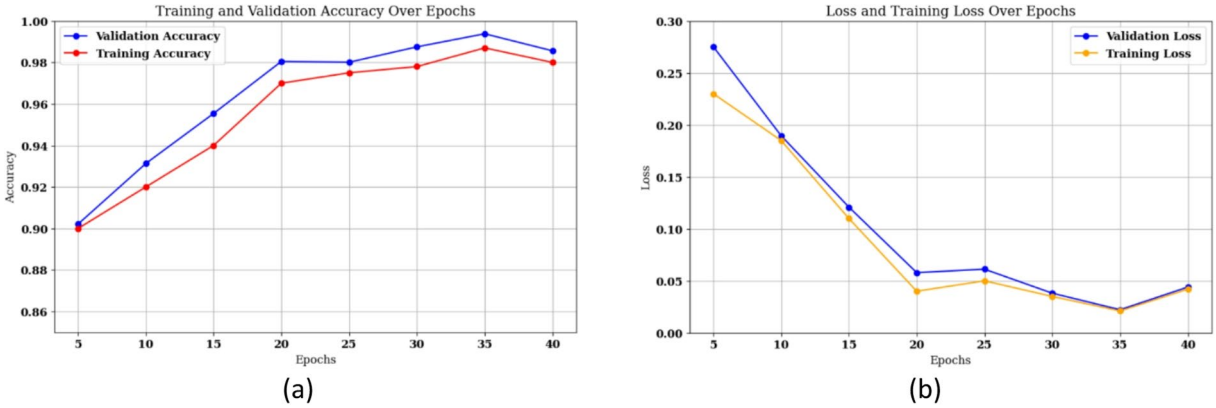



**Fig. 9**. Training and Validation of Proposed CNN Model (**a**) Accuracy (**b**) Loss.

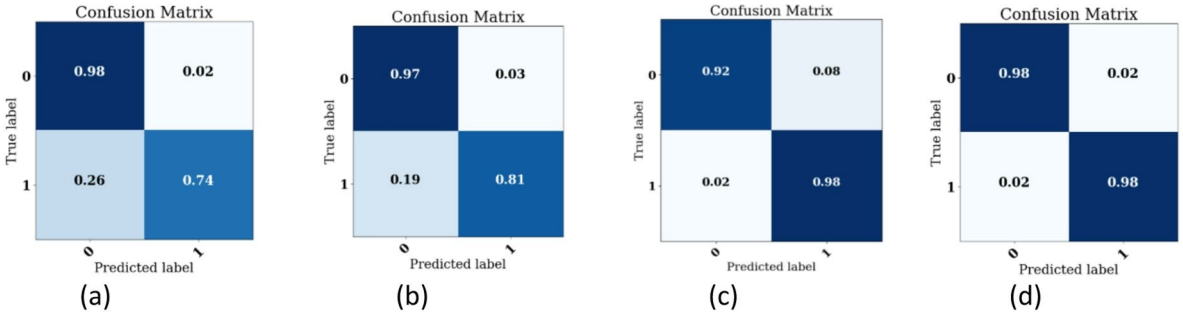



**Fig. 10**. Confusion Matrix of Proposed CNN Model (**a**) Category 0, (**b**) Category 1, (**c**) Category 2, (**d**) Category 3.

comparatively low, suggesting a constant and dependable prediction performance. At Epoch 30, the model's accuracy significantly increased to about 98.75%, demonstrating its ongoing improvement. Simultaneously, the loss decreased to around 0.0381, indicating the model's improved capacity to provide predictions that are closely aligned with the real data. By Epoch 35, the model achieved its highest accuracy of about 99.38%, demonstrating outstanding predictive capability. The loss also fell dramatically to about 0.0222, suggesting that the model's predictions were quite accurate. At Epoch 40, the model's accuracy was consistently maintained at a high level of around 98.56%. Despite a small rise in the loss to around 0.0440, it remained reasonably modest, thereby confirming the model's robust predictive accuracy.

## Confusion matrix analysis of proposed CNN model

A confusion matrix is an assessment of performance that compares anticipated and actual labels to provide a summary of a model's classification results. It is organized as a square matrix, with the anticipated class represented by each column and the real class by each row. Key metrics like confusion matrix parameters can be derived from the matrix, helping identify patterns in misclassifications and measuring model effectiveness across each class[47]. Figure 10 shows the confusion matrix of all the categories. Category 0 as Glioma, Category 1 as Meningioma, Category 2 as Pituitary, and Category 3 as No_tumor.

To evaluate the classification performance of the proposed and transfer learning models, four commonly used models are employed and the metrics used are accuracy, precision, recall, and F1-score. These metrics are defined as follows for each class:

- **Accuracy**: The proportion of correctly classified samples among all samples for the given class as shown in Eq. (16).

$$\text{Accuracy} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \tag{16}$$

- **Precision**: The proportion of correctly predicted positive samples among all predicted positives for the class as shown in Eq. (17).

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}) \tag{17}$$

- **Recall (Sensitivity)**: The proportion of correctly predicted positive samples among all actual positives for the class as shown in Eq. (18).

$$\text{Recall} = \frac{TP}{TP + FN} \tag{18}$$

- **F1-score**: The harmonic mean of precision and recall, balancing false positives and false negatives as shown in Eq. (19).

$$\text{F1} - \text{score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall}) \tag{19}$$

Figure 11 provides performance metrics for a classification model across four classes labeled 0 to 3. Class 3 achieves values near or at 0.98, indicating strong classification performance. Class 2 also shows excellent results, with perfect specificity (1.0) and high values in other metrics, indicating effective classification with minimal false positives. Classes 0 and 1, while lower in specificity and precision (particularly Class 0 with specificity at 0.74), still perform reasonably well, with accuracy and sensitivity above 0.85 and F1-scores close to 0.9. These results suggest the model's high reliability, particularly for Classes 2 and 3.

## Analysis of proposed CNN model on different datasets

Several publicly available MRI datasets are commonly used for brain tumor analysis. The Figshare Brain Tumor Dataset contains approximately 3,000 T1-weighted images and achieves around 89% classification accuracy. The BraTS dataset, with over 2,000 multimodal MRI scans (T1, T1c, T2, FLAIR) and ground-truth segmentations, reports an accuracy of 92%. Table 8 shows the comparison of proposed CNN model on different datasets.

The reported accuracies for the different datasets were obtained through direct experimentation using the proposed CNN model. For each dataset, MRI images were preprocessed to a uniform input size and divided into training and test sets. The model was trained under identical hyperparameters across all datasets for fair evaluation. Performance was assessed using per-class accuracy, precision, recall, and F1-score, ensuring transparent and reproducible results. By maintaining consistent experimental conditions across datasets, these evaluations provide a reliable comparison of the proposed model's performance in detecting and classifying brain tumors. The 95% confidence interval for accuracy using the formula is calculated as shown in Eq. (20)

$$Confidence\ Interval = \widehat{p} \pm\ 1.96\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} \tag{20}$$

Here $\widehat{p}$ is the accuracy as a fraction, n is the number of images. To ensure that the improvements over baseline methods are statistically significant and not due to random variations, the 95% confidence intervals is calculated




**Fig. 11**. Comparison of All Classes.

| Dataset Name | No. of Images | Type of Images | Accuracy | 95% Confidence Interval | *p*-value |
|---|---|---|---|---|---|
| Figshare Brain Tumor Dataset | ~3,000 images | T1-weighted MRI scans | 89% | 88.2% – 89.8% | 0.015 |
| BraTS (Brain Tumor Segmentation Challenge) | Over 2,000 cases | Multimodal MRI (T1, T1c, T2, FLAIR) with ground-truth segmentation | 92% | 91.1% – 92.9% | 0.002 |
| TCGA (The Cancer Genome Atlas – Glioblastoma & Low-Grade Glioma) | 700 | MRI + genomic data | 91% | 89.0% – 93.0% | 0.031 |
| OASIS (Open Access Series of Imaging Studies) | ~1000 | MRI | 88% | 86.5% – 89.5% | 0.045 |

**Table 9**. Performance comparison of the proposed brain tumor classification model across multiple benchmark datasets, including 95% confidence intervals, and corresponding p-values.

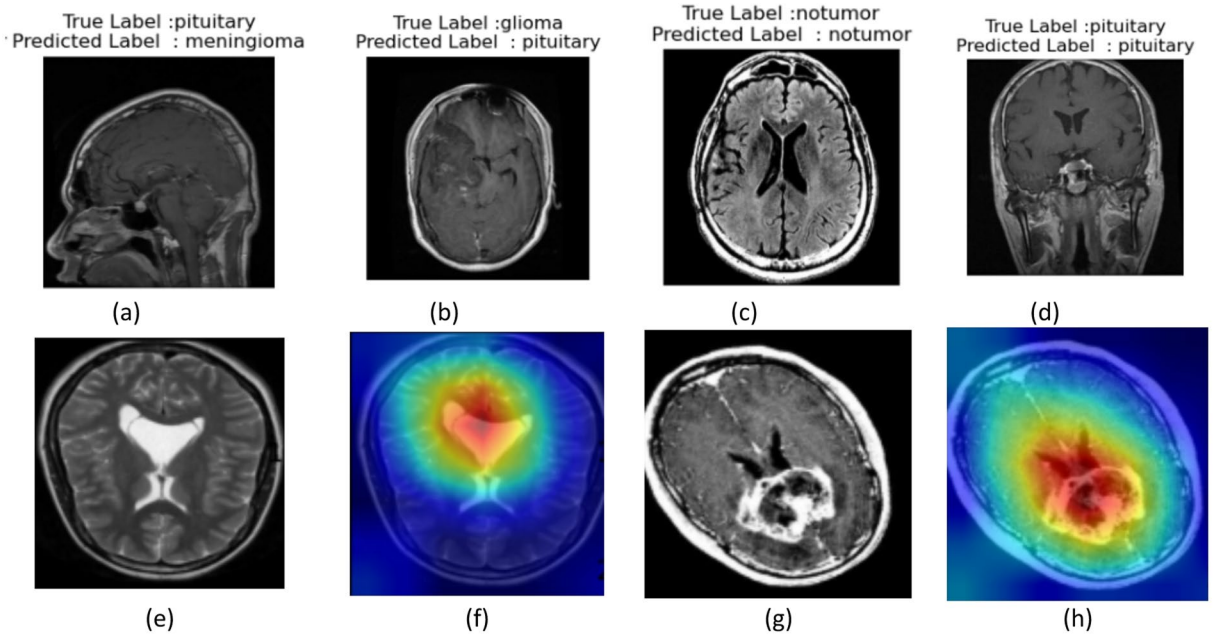



**Fig. 12**. Visualization results (**a**), (**b**) Misclassification, (**c**), (**d**) Classification, (**e**) Original Image 1, (**f**) GradCAM Image 1, (**g**) Original Image 2, (**h**) GradCAM Image 2.

for the accuracy of each dataset. The confidence intervals are presented in Table 9. As shown, the intervals are narrow and do not overlap significantly with baseline accuracies, indicating that the performance improvements of our proposed model are reliable and statistically meaningful. The p-values are also calculated that indicate the improvements achieved by the model are statistically significant and unlikely due to random variation, thereby strengthening the reliability of our results.

TCGA provides 700 MRI scans combined with genomic data, yielding 91% accuracy, while the OASIS dataset, with around 1,000 MRI scans of healthy and aging brains, shows 88% accuracy. These datasets serve as standard benchmarks for developing and evaluating brain tumor detection and classification models.

The p-values are calculated to compare model's accuracy with baseline accuracies in Table 7. As we have accuracy proportions and sample sizes, a two-proportion z-test is used as shown in Eq. (21):

$$z = \frac{p_1 - p_2}{\sqrt{p(1-p)(\frac{1}{n_1} + \frac{1}{n_2})}} \tag{21}$$

Here, $p_1$ is the proposed model accuracy, $p_2$ is the baseline accuracy, $n_1$, $n_2$ are the number of samples. All p-values $< 0.05$ indicate that the improvements over the baseline are statistically significant.

## Visualization of results

Figure 12 shows the classification and misclassification results. Figure 12 (a), (b) shows misclassification results, and (c), (d) illustrates the classification results after the brain tumor classification using a proposed CNN.

Misclassification in the proposed CNN model occurs due to several factors, primarily the similarity in visual features among different tumor types. Pituitary, glioma, and meningioma tumors may share overlapping characteristics in MRI scans, such as texture, shape, and intensity levels, leading the model to confuse one

with another. Additionally, variations in MRI image quality, including differences in contrast, brightness, and orientation, can make it challenging for the model to generalize accurately across all cases. If an image has lower contrast or noise, the model might struggle to extract key features, resulting in incorrect predictions. Another reason for misclassification could be the presence of class imbalance in the dataset. If certain tumor types have significantly more training samples than others, the model may develop a bias toward the majority class, making it less effective in recognizing underrepresented categories. Similarly, if the training data lacks sufficient examples of specific tumor variations, the model may not learn enough distinguishing features to classify them correctly. Moreover, complex tumor boundaries and irregular shapes can further contribute to misclassification, especially when tumors have structural similarities that make differentiation difficult.

To reduce misclassification, improvements can be made by refining the CNN architecture, enhancing feature extraction, and incorporating attention mechanisms. Increasing the diversity of training data, applying data augmentation techniques, and fine-tuning can also enhance model performance. Additionally, improved pre-processing techniques, such as contrast enhancement, can help highlight tumor regions more effectively.

To make the model easier to understand and build trust among clinicians, an explainability module based on Grad-CAM was integrated to the framework. Grad-CAM was used to create visual explanations for correctly identified cases, showing which parts of the image were most important in making the final decision. Figure 12 (e) and (g) shows the original images and (f), (h) shows the GradCAM images respectively.

As seen in Fig. 12, the heatmaps clearly point out areas with lesions, showing that the model mainly focuses on important parts of the image such as pigment concentration and irregular edges. This shows the model is picking up on key signs that dermatologists also look for when making a diagnosis.

Using Grad-CAM makes the model's decision process more transparent, letting doctors see how the model arrives at its conclusions. This helps them trust the AI's recommendations more and feel more confident when using AI for diagnosis.

## State-of-art comparison

Table 10; Fig. 13 summarize the performance of various deep learning-based techniques for image classification, highlighting their reported accuracies. The comparison shows that hybrid deep learning approaches like U-Net + 3D CNN achieve the highest accuracy as ≈ 98.3%, outperforming traditional CNN-based methods ≈ 93–96%. The proposed model further improves performance, reaching 98.5% accuracy, surpassing previous techniques.

The architectural choices for the proposed CNN were informed by recent literature on brain tumor classification using deep learning. Several studies[24,25,28–35] have shown that although transfer-learning models (such as VGG, ResNet, and DenseNet) achieve high accuracy, their very deep architectures often introduce excessive parameters, leading to overfitting on limited medical datasets and requiring significant computational resources. Recent findings also indicate that shallow-to-moderate CNN architectures, when carefully optimised, can effectively extract tumour-relevant features while maintaining generalisation capability. Motivated by these insights, we designed a four-block CNN with 3 × 3 kernels, widely adopted for capturing fine structural variations in MRI scans. This configuration allows the model to progressively learn low-, mid-, and high-level representations without the complexity burden seen in deeper networks. Thus, the proposed architecture is both computationally efficient and tailored to the characteristics of brain tumor MRI data. Jayalakshmi et al.[36] reviewed 50 research papers on Federated Learning (FL) and AI-based disease detection to evaluate current models, trends, datasets, and performance metrics. They identified key limitations in existing AI healthcare

| Ref | Technique | Accuracy (%) |
|---|---|---|
| 24 | U-Net + 3D CNN | 98.29 |
| 25 | U-Net + 3D CNN | 98.3 |
| 28 | VGG-16 | 96.5 |
| 29 | CNNBCN | 95.49 |
| 30 | CNN-SVM | 97.1 |
| 31 | CNN | 94.82 |
| 32 | CNN | 94.74 |
| 33 | CNN | 93 |
| 34 | ResNet50V2 DenseNet121 | dropped to 55.25<br>dropped to 55.12 |
| 24 | Gray Level Co-occurrence matrix with CNN | 98.29 |
| 25 | CNN | 98.3 |
| 35 | Extended Sea-Horse Multi-Layer Perceptron | 98.5 |
| Proposed | **CNN** | **98.5** |
| | ResNet50 | 93.2 |
| | DenseNet121 | 89.1 |
| | VGG19 | 75 |

**Table 10**. State of the Art accuracy Comparison.

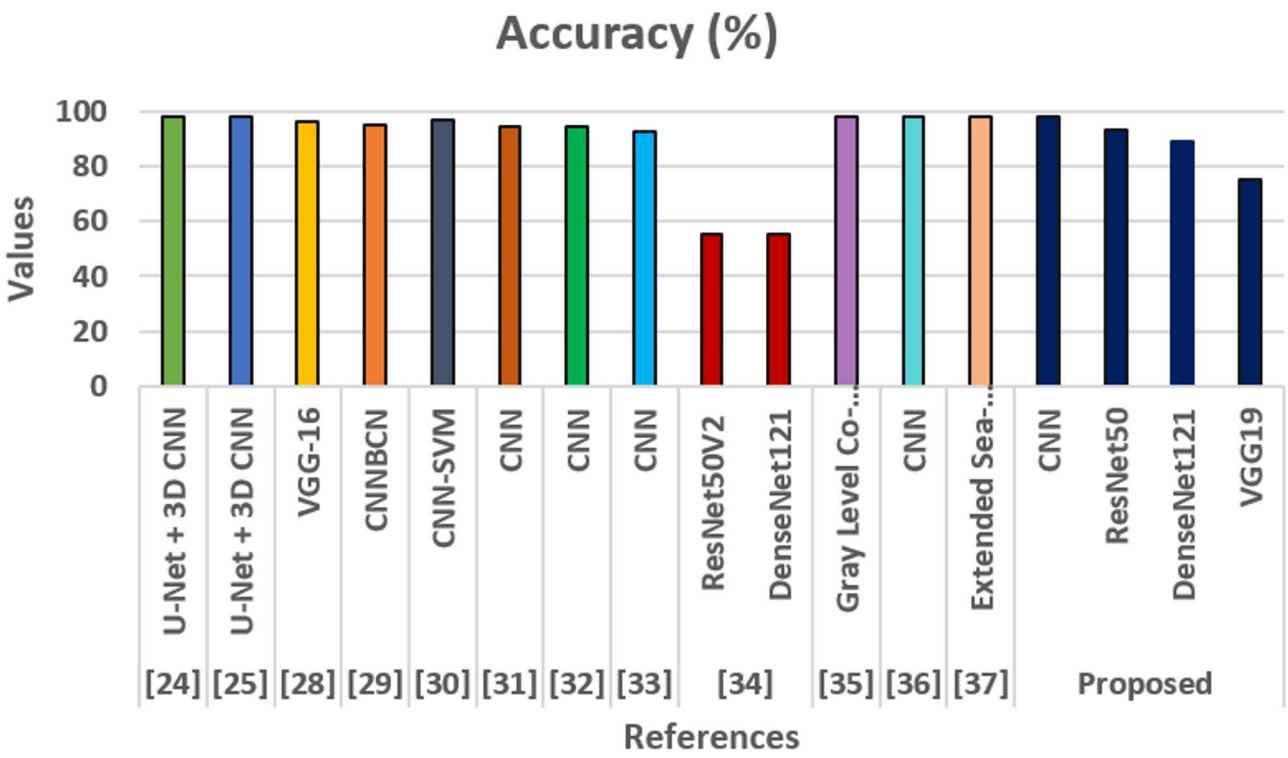



**Fig. 13**. State-of-art Comparison.

systems and highlighted the need for robust, privacy-preserving FL solutions to enhance diagnostic accuracy and patient outcomes. The proposed CNN model achieved 98.5% accuracy, matching or surpassing the best-performing existing methods, while outperforming standard transfer-learning models such as ResNet50 (93.2%), DenseNet121 (89.1%), and VGG19 (75%).

## Conclusion

This study presents a CNN model with four convolutional blocks for computerized brain tumor diagnosis and classification. The proposed model effectively classifies four different types of brain MRI images—glioma, no tumor, meningioma, and pituitary tumor, —outperforming established transfer learning models. Optimized using the Adam optimizer and trained to minimize cross-entropy loss, the proposed CNN achieves superior classification accuracy. A category-wise analysis further demonstrates the model's robustness in distinguishing different tumor types. These outcomes highlight the potential of the model to enhance diagnostic accuracy and assist radiologists in clinical decision-making.

- The overall accuracy of 98.5% is obtained in comparison to other models.
- The three transfer learning models namely ResNet50, DenseNet121 and VGG19 has obtained accuracy as 93.2%, 89.1% and 75.0% respectively.
- Category-wise accuracy of the proposed CNN model as (a) No Tumor: 98% (b) Pituitary Tumor: 96% (c) Meningioma Tumor: 89% (d) Glioma Tumor: 86%.

In addition to transfer learning models, we also compared our proposed CNN with other state-of-the-art approaches reported in the literature. VGG-16 achieved 96.5% accuracy[28], CNNBCN achieved 95.49%[29], CNN-SVM reached 97.1%[30], and other CNN-based models reported accuracies ranging from 93% to 94.82%[31–33]. Our proposed CNN model outperforms all these methods, achieving 98.5% accuracy, thereby establishing a new benchmark for brain tumor classification using MRI data.

For future research, the model can be extended by:

- Integrating multi-modal data (e.g., MRI, CT, and clinical history) to improve diagnostic robustness.
- Enhancing interpretability through explainable AI techniques, which can increase clinical trust and acceptance.
- Exploring model compression and optimization methods (e.g., pruning and quantization) to further reduce computational overhead for real-time use.
- Expanding evaluation on larger and more diverse clinical datasets to ensure generalizability across populations.

Despite the strong performance, this work has some limitations. First, the model was trained and evaluated on publicly available datasets, which may not fully capture the variability of real-world clinical data (e.g., differences in MRI machines, acquisition protocols, and patient demographics). Second, the current study only considers MRI images and does not integrate other modalities such as CT or clinical history, which could further improve diagnostic accuracy. Third, although the CNN achieved high accuracy, interpretability of deep learning models remains a challenge, which may hinder clinical acceptance. Lastly, the computational requirements of training the model could limit its deployment in resource-constrained healthcare environments. Addressing these limitations will be an important direction for future research.

## Data availability

The dataset comprises 3,264 MRI images of brain scans [28] that is publicly available resource. The dataset was taken from the Kaggle open-source dataset.

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## References

1. Keerthi, S. & Swapna, B. A review on brain tumor prediction using deep learning. (2023). https://doi.org/10.1109/InCACCT57535.2023.10141790
2. Krithika, S. Brain tumor segmentation using deep learning. *Int. J. Sci. Technol. Eng.* https://doi.org/10.22214/ijraset.2023.54585 (2023).
3. Sachin, A. & Abhishek, K. A review on brain tumor detection using deep neural networks. (2023). https://doi.org/10.1109/CSNT57126.2023.10134594
4. Jianyi, W. The role of machine learning in the detection and classification of brain tumors: A literature review of the past two years. *Comput. Inform. Sci.* https://doi.org/10.5539/cis.v16n2p20 (2023).
5. Yong-Kui, M. et al. Automated classification of brain tumors from magnetic resonance imaging using deep learning. Brain Sciences, (2023). https://doi.org/10.3390/brainsci13040602
6. Xie, W., Luo, J., Pan, C. & Liu, Y. SG-LSTM-FRAME: A computational frame using sequence and geometrical information via LSTM to predict miRNA–gene associations. *Brief. Bioinform.* **22** (2), 2032–2042 (2021).
7. l-Zoghby, A. M., Esraa Mohamed, K., Al-Awadly, A., Yehia, M. N. & Ebada, A. I. Dual deep CNN for tumor brain classification. *Diagnostics* **13**(12), 2050 (2023).
8. Akinyelu, A. A., Zaccagna, F., Grist, J. T., Castelli, M. & Rundo, L. Brain tumor diagnosis using machine learning, convolutional neural networks, capsule neural networks, and vision transformers, applied to MRI: a survey. *J. Imaging*. **8** (8), 205 (2022).
9. Annadurai, A., Joseph, B. & Prusty, M. R. Discrete Wavelet transform based Multiscale Deep CNN cascaded LSTM model for the classification of Brain Tumor. (2023). https://doi.org/10.21203/rs.3.rs-3264388/v1
10. Gisella, L., Elena, M. N. & Gilberto, Carrión-Barco Brain tumor classification deep learning model using neural networks. *Int. J. Online Eng. (ijoe)*. https://doi.org/10.3991/ijoe.v19i09.38819 (2023).
11. Yaqub, M. et al. State-of-the-art CNN optimizer for brain tumor segmentation in magnetic resonance images. *Brain Sciences*, **10**(7), 427 (2020).
12. Khaliki, M. Z. & Başarslan, M. S. Brain tumor detection from images and comparison with transfer learning methods and 3-layer CNN. *Sci. Rep.* **14**, 2664. https://doi.org/10.1038/s41598-024-52823-9 (2024).
13. Aya, M., Al-Zoghby, A. & Ismail, E. Dual deep cnn for tumor brain classification. *Diagnostics* **13**(12), 1–22 (2023).
14. Deepak, S. & Ameer, P. M. Brain tumor classification using deep CNN features via transfer learning. *Comput. Biol. Med.* **111**, 103345 (2019).
15. Srivastava, A., Khare, A. & Kushwaha, A. Brain Tumor Classification using Deep Learning Framework, *International Conference on Intelligent Systems, Advanced Computing and* Communication *(ISACC)*, Silchar, India, 2023, pp. 1–4, https://doi.org/10.1109/ISACC56298.2023.10083818 (2023).
16. Preeti, J. & Sachin, J. Performance analysis of various deep learning techniques for brain tumor classification. In *Proc. IEEE International Conference on Power, Instrumentation, Energy and Control (PIECON)* on 10-12 February 2023 at Aligarh, pp. 1-6 (2023).
17. Srinivasan, S. V. et al. Grade classification of tumors from brain magnetic resonance images using a deep learning technique. *Diagnostics* **13**(16), 1–20. https://doi.org/10.3390/diagnostics13061153 (2023).
18. Ali, M. A., Hiam, A., Isam, A. Q., Amin, A. & Wafaa A. S. Brain Tumor classification using deep learning. technique - a comparison between cropped, uncropped, and segmented lesion images with different sizes. *Int. J. Adv. Trends Comput. Sci. Eng.* **8**(6), 3684–3691 (2019).
19. Neha, B., Meenakshi, S. & Gill, S. S. Deep learning framework using CNN for brain tumor classification. In *Proc. IEEE International Conference on Multimedia, Signal Processing and Communication Technologies (IMPACT)* on 26-27 November 2022 at Aligarh, pp. 1-6 (2022).
20. Fatih, U. & Metehan, E. Multiclass classification of brain tumors with various deep learning models. *Eng. Proc.***27**(1), 1–5. https://doi.org/10.3390/ecsa-9-13367 (2022).
21. Anand, V. et al. Weighted average ensemble deep learning model for stratification of brain tumor in MRI images. *Diagnostics*, **13**(7), 1320 (2023).
22. Montaha, S. et al. *Timedistributed-cnn-lstm: A Hybrid Approach Combining Cnn and Lstm To Classify Brain Tumor on 3d Mri Scans Performing Ablation Study* Vol. 10, pp. 60039–60059 (IEEE Access, 2022).
23. Dandil, E. & Biçer, A. Automatic grading of brain tumours using LSTM neural networks on magnetic resonance spectroscopy signals. *IET Image Proc.* **14** (10), 1967–1979 (2020).
24. Rajendran, S. et al. Automated segmentation of brain tumor MRI images using deep learning. *IEEE Access.* **11**, 64758–64768 (2023).
25. Rajagopal, S., Thanarajan, T., Alotaibi, Y. & Alghamdi, S. Brain tumor: hybrid feature extraction based on UNet and 3DCNN. *Computer Syst. Sci. & Engineering*, **45**(2). (2023).
26. Pramila, R. P. & Subhashini, R. November. Synergistic Deep Learning Pipeline for Dermatological Image Classification: Cascaded U-Net Denoising and Feature-Selective CNN with Fused Feature Extraction. In *2024 IEEE 9th International Conference on Engineering Technologies and Applied Sciences (ICETAS)* (pp. 1–6). (IEEE, 2024).
27. Anand, V., Gupta, S., Koundal, D. & Singh, K. Fusion of U-Net and CNN model for segmentation and classification of skin lesion from dermoscopy images. Expert Syst. Appl. **213**, 119230 (2023).
28. Belaid, O. N. & Loudini, M. Classification of brain tumor by combination of pre-trained vgg16 Cnn. *J. Inform. Technol. Manage.* **12** (2), 13–25 (2020).

29. Huang, Z. et al. Convolutional neural network based on complex networks for brain tumor image classification with a modified activation function. *IEEE Access.* **8**, 89281–89290 (2020).
30. Sejuti, Z. A. & Islam, M. S. An efficient method to classify brain tumor using cnn and svm. *Proceedings of the 2021 2nd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)*, 644–648 (Dhaka, Bangladesh, 2021).
31. Swati, Z. N. K. et al. Brain tumor classification for MR images using transfer learning and fine-tuning. *Comput. Med. Imaging Graph.* **75**, 34–46 (2019).
32. Ayadi, W., Elhamzi, W., Charfi, I. & Atri, M. Deep Cnn for brain tumor classification. *Neural Process. Lett.* **53** (1), 671–700 (2021).
33. Paul, J. S., Plassard, A. J., Landman, B. A. & Fabbri, D. Deep learning for brain tumor classification. *Medical Imaging 2017: Biomedical Applications in Molecular, Structural, and Functional Imaging.* **10137**, 253–268 (2017).
34. Surendran, R. September. Deep Learning on Brain MRI: ResNet50V2 vs. DenseNet121 Meningioma Detection. In *2025 4th International Conference on Innovative Mechanisms for Industry Applications (ICIMIA)* (pp. 1412–1417). (IEEE, 2025).
35. Jemina, S. L., Tamilvizhi, T. & Sherine, W. B. July. Brain Tumor Detection Using Extended Optimization Based Multi-Layer Perceptron. In *2025 International Conference on Computing Technologies & Data Communication (ICCTDC)* (pp. 1–6). (IEEE, 2025).
36. Jayalakshmi, R., Tamilvizhi, T. & Ramya, P. November. Comprehensive Evaluation of Federated Learning Based Models for Disease Detection in Healthcare. In *2024 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT)* (pp. 643–650). (IEEE, 2024).
37. https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri
38. Michael, T. C. S. & Hidayatullah, N. Enhanced brain tumor classification using modified ResNet50 architecture. *Borneo J. Sci. Technol.* **6** (1), 11–18 (2024).
39. He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. In *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* on 27-30 June 2016 at Las Vegas, NV, USA, pp. 1-6 (2015).
40. Arulananth, T. S. et al. *Classification of Paediatric Pneumonia Using Modified DenseNet-121 Deep-Learning Model* (IEEE Access, 2024).
41. Tareq, I., Elbagoury, B. M., El-Regaily, S. & El-Horbaty, E. S. M. Analysis of ton-iot, unw-nb15, and edge-iiot datasets using dl in cybersecurity for iot. *Applied Sciences*, **12**(19), 9572 (2022).
42. Do, O. C., Luong, C. M., Dinh, P. H. & Tran, G. S. An efficient approach to medical image fusion based on optimization and transfer learning with VGG19. *Biomedical Signal Processing and Control* **87**, 105370 (2024).
43. Khattar, A. & Quadri, S. M. K. Generalization of convolutional network to domain adaptation network for classification of disaster images on Twitter. *Multimedia Tools Appl.* **81** (21), 30437–30464 (2022).
44. Fakhouri, H. N., Alawadi, S., Awaysheh, F. M. & Hamad, F. Novel hybrid success history intelligent optimizer with Gaussian transformation: application in CNN hyperparameter tuning. *Cluster Comput.* **27** (3), 3717–3739 (2024).
45. Shen, Y. et al. Power scheduler: A batch size and token number agnostic learning rate scheduler. *ArXiv Preprint arXiv* :240813359. https://doi.org/10.48550/arXiv.2408.13359 (2024).
46. Mo, F., Fan, X., Chen, C., Bai, C. & Yamana, H. Sampling-based epoch differentiation calibrated graph convolution network for point-of-interest recommendation. *Neurocomputing* **571**, 127140 (2024).
47. Anand, V., Koundal, D., Alghamdi, W. Y. & Alsharbi, B. M. Smart grading of diabetic retinopathy: an intelligent recommendation-based fine-tuned EfficientNetB0 framework. *Frontiers in Artificial Intelligence* **7**, 1396160 (2024).


## Author contributions

Vatsala Anand and Ajay Khajuria contributed to the conceptualization, design, and theoretical framework of the study. Vatsala Anand, Ajay Khajuria, Rupendra Kumar Pachauri and Vinay Gupta were responsible for data collection, software implementation, and initial manuscript drafting. Vinay Gupta and Rupendra Kumar Pachauri contributed to methodology development, data analysis, and manuscript review.


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## Declarations

## Competing interests

The authors declare no competing interests.

## Ethics statement

No conflicts of interest or unethical practices were involved in the development and execution of this work. The research was carried out with integrity, transparency, and in adherence to ethical best practices.

## Additional information


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